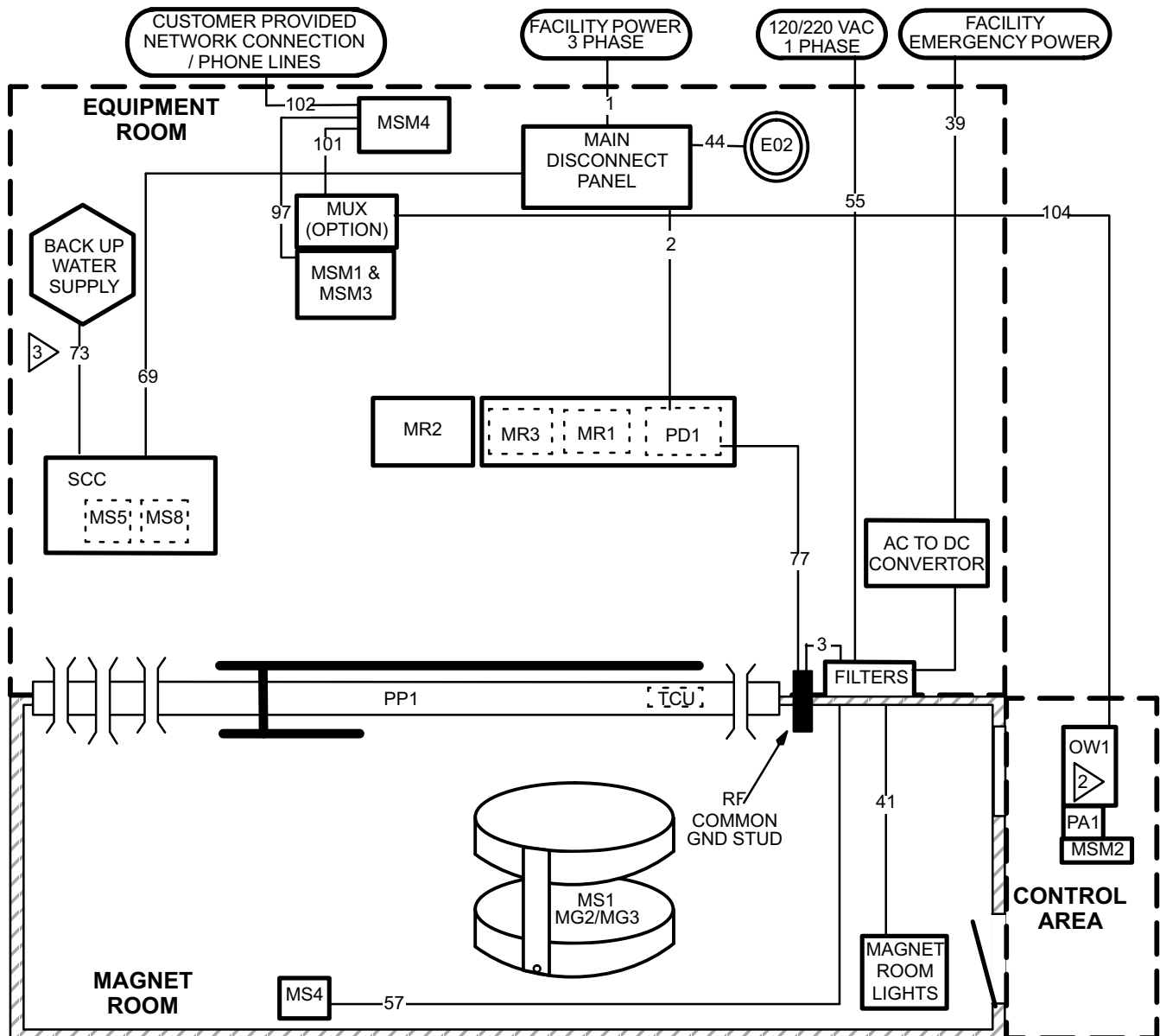


6-4-8 Customer Supplied Interconnects

Illustration 6-8 shows the cable Groups which are customer supplied. Table 6-9 contains the list of customer supplied interconnects for the MR system and references to location of detail information for the specific interconnects.



- 2 OPERATOR WORKSPACE (OW1) SUBSYSTEM EQUIPMENT IS PROVIDED WITH MAXIMUM LENGTH CABLES POSSIBLE. SEVERAL OW ASSEMBLIES ARE LOCATED ON THE OW TABLE & OW INTERCONNECTS ARE ROUTED THROUGH TABLE CABLE TRAY. FOR REFERENCE USE ONLY THE OW1 RUN NUMBERS FOR LINUX PC ARE 1059, 1060, 1061, 1062, 1063, 1064, 1065, 1066, 1067, 1068, 1069, 1070, 1071, 1074, 1075.
- 3 THIS GROUP CONTAINS WATER LINES WHICH SHALL BE ROUTED SEPARATE FROM ELECTRICAL LINES (I.E. POWER & SIGNAL).

SYSTEM CUSTOMER SUPPLIED INTERCONNECTS DIAGRAM

ILLUSTRATION 6-9

6-4-8 Customer Supplied (Continued)

TABLE 6-9
CUSTOMER SUPPLIED INTERCONNECTS

GROUP NUMBER	GROUP AREA in. ² mm ²)	BETWEEN UNITS		RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO			FROM	TO	
1	—	Facility Power	MDP	—	—	—	—	Customer supplied, See Note 1.
2	—	MDP	PD1 See Note 2	—	—	—	—	Customer supplied. Refer to Section 5-3-2 System Power Distribution Unit for wire size information.
3	—	Facility Emerg Power Filter	PP1	—	—	—	—	Customer supplied Ground.
39	—	Facility Emerg Power	Filter	—	—	—	—	Refer to Section 5-8 for DC Lighting Controller cabling.
41	—	Filter	Magnet Room Lights	—	—	—	—	Refer to Sections 5-7 and 5-8 .
44	—	MDP	EO2	—	—	—	—	Customer supplied Refer to Section 6-3 .
55	—	Facility Power	Filter	—	—	—	—	Customer supplied Magnet Room power (refer to Sections 5-1 and 7-4).
57	—	Filter	MS4	—	—	—	—	Customer supplied (refer to Sections 5-1).
69	—	MDP	SCC	—	—	—	—	Customer Furnished.
73 See Note 3	—	Backup Water	SCC	—	—	—	—	Customer Furnished water lines (refer to Section 4-4-1 Cryo Cooler Compressor Backup Temporary Water Cooling Recommendation).
77	—	PD1 See Note 2	RF COM-MON GND STUD	—	—	Ring Terminal	Ring Terminal	Customer supplied ground cable, refer to Section 5-4-2 System Ground .
97	0.07 (45.2)	MSM1 & MSM3	MSM4	—	0.30 (7.62)	1.30x2.00 (33.5x50.8)	1.30x2.00 (33.5x50.8)	Customer provided phone line, cable diameter & plug pull information are estimates.
Note 1 If low Voltage Step-Up Transformer Option (R4500AS or R4500BD) is used then customer supplied interconnects are required between facility power, transformer and MDP. 2 The PDU is a module (PD1) in the lower portion of the Power Cabinet (MR1). 3 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).								
(Continued)								

6-4-8 Customer Supplied (Continued)

TABLE 6-9
CUSTOMER SUPPLIED INTERCONNECTS

GROUP NUMBER	GROUP AREA in. ² mm ²)	BETWEEN UNITS		RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
		FROM	TO			FROM	TO	
101	—	MSM4	MUX	—	—	—	—	Customer provided phone line. Phone line routed through UPS for transient protection.
102	—	Phone Line Con- nection	MSM4	—	—	—	—	Refer to Section 2-7-5 System Monitoring & Support Connectivity Requirements for additional customer phone line information.
104	—	MUX	OW1 InSite Modem	—	—	—	—	Customer provided phone line.

6-4-9 System Cooling Cabinet With Remote Condenser Unit Additional Interconnects

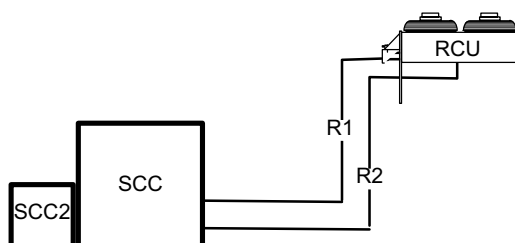
The SCC with RCU requires the interconnects defined in this section in addition to the interconnects in Section 6-4 SYSTEM INTERCONNECTS.

Installation of the RCU must be in accordance with local and national codes for electrical and according to standard accepted refrigeration practices (i.e. proper traps in lines). The following recommendations should be adhered to:

- Use only refrigeration grade copper tubing.
- Soft solder joints are not acceptable
- Put dry nitrogen through lines while brazing.
- Do not leave dehydrated piping or components open to the atmosphere any longer than absolutely necessary.

Note

Refer to **Section 2-7-1 System Cooling Cabinet Subsystem Equipment** Table 2-5 for listing of responsibility for the specific installation tasks. Also refer to Ellis & Watts *Technical Publication 468 Pre installation/Installation/Operating LC 20M/RAC Chilled Water System for GE Signa OpenSpeed MRI Equipment* for additional SCC with RCU information and details.



REMOTE CONDENSER UNIT & SYSTEM COOLING CABINET CUSTOMER PROVIDED INTERCONNECTS
ILLUSTRATION 6-10

TABLE 6-10
CUSTOMER PROVIDED REMOTE CONDENSER UNIT INTERCONNECT LIST

GROUP NUMBER	GROUP AREA in. ² (mm ²)	BETWEEN UNITS		USABLE LENGTH ft (m)	RUN NUMBER	CABLE DIAMETER in. (mm)	PLUG PULLING DIAMETER X LENGTH in. (mm)		NOTES
R1	1.595* (1029)*	SCC	RCU	See Note 1	gas	1.125* (28.6)*	refrigeration tubing	refrigeration tubing	RCU is supplied with sweat fittings on the SCC and RCU. Refer to Illustrations 2-15 and 2-16. Refrigeration tubing is Customer Supplied.
					liquid	0.875* (22.2)	refrigeration tubing	refrigeration tubing	
R2		SCC	RCU	See Note 1	power	0.064 See Note 2	hard wired	hard wired	At RCU Customer Supplied power wiring is connected to Lock Out /Tag Out Breaker Box supplied and installed on the unit.
Note * Dimension values are estimated, need to determine dimensions for customer supplied materials. 1 The SCC and RCU must not be separated by a distance greater than 200 ft (61 m) of customer provided refrigeration tubing. The vertical separation must not exceed 100 ft (30.5 meter) for the RCU above the SCC and 10 ft (3 m) if for the RCU is below the SCC. 2 Customer provided power wire from SCC to RCU must be 14 AWG (25 mm ²) minimum with actual wire size to be determined by customer/contractor dependent on total run length. RCU power demands for are Start-up 10 Amps and Full Load 3.7 Amps for 2 fans (1@1/3 hp 1.3 FLA and 1@3/4 hp 2.4 FLA). Refer to vendor manual for additional information .									

6-4-10 Water Cooled System Cooling Cabinet Additional Interconnects

The water cooled SCC requires the customer provided water cooling lines, refer to Section 4-4-2 Water Cooled System Cooling Cabinet Configuration Requirements for details.

Note

Water lines shall be routed separate from electrical lines (i.e. power & signal).

6-4-11 Cable Group & Location Cross Reference

Cable groups shown in Illustration 6-1 are listed in Table 6-11 which contains the Location for each Group and a reference to the subsection or table which has the group content details and the usable length provided by the listed Site Collector catalogs.

TABLE 6-11
LOCATION & CABLE GROUP CROSS REFERENCE TO RUNS DETAILS

CABLE GROUP #	BETWEEN UNITS		LOCATION See Note 1 for definition of L numbers	SECTION OR TABLE WITH GROUP CONTENT DETAILS
	FROM	TO		
1	Facility Power	MDP	Customer Supplied	Section 6-4-8 Table 6-9
2	MDP	PD1	Customer Supplied	Section 6-4-8 Table 6-9
3	Facility Emerg Power Filter	PP1	Customer Supplied	Section 6-4-8 Table 6-9
4 to 5	—	—	Group Not Used	—
6	PD1	MR2	L3	Section 6-4-5 Table 6-6
7 to 19	—	—	Group Not Used	—
20	MR2	MR3 in MR1	L3	Section 6-4-5 Table 6-6
21	—	—	Group Not Used	—
22	MR2	MR1	L3	Section 6-4-5 Table 6-6
23	MR2	PP1	L2	Section 6-4-3 Table 6-4
24	—	—	Group Not Used	—
25	MR1	PP1	L2	Section 6-4-3 Table 6-4
26	MR1	MG2/3	L1/L2	Section 6-4-4 Table 6-5
27	MR3	MG2	L1/L2	Section 6-4-4 Table 6-5
28 to 38	—	—	Group Not Used	—
39	Facility Emerg Power	Filter	Customer Supplied	Refer to Section 5-8 for DC Lighting Controller cabling.
40	—	—	Group Not Used	—
41	Filter	Magnet Room Lights	Customer Supplied	Refer to Sections 5-7 and 5-8 .
42	PP1	EO1	L1	Section 6-4-2 Table 6-3
43	MR2	RF Door Switch	L5	Section 6-4-7 Table 6-8
44	MDP	E02	Customer Supplied	Section 6-4-8 Table 6-9
45	PP1	MG2/3	L1	Section 6-4-2 Table 6-3
Note 1 Interconnects LOCATION is defined as the following: L1 Within Magnet Room between components L2 Between Penetration Panel (PP1) and components in Equipment Room L1/L2 Between Magnet Room and Equipment Room components through PP1 L3 Between components within Equipment Room L4 Between Operator Workspace (OW) and PP1 L5 Between OW and components in Equipment Room 2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).				
(Continued)				

6-4-11 Cable Group & Location Cross Reference (Continued)

TABLE 6-11 (Continued)
LOCATION & CABLE GROUP CROSS REFERENCE TO RUNS DETAILS

CABLE GROUP #	BETWEEN UNITS		LOCATION See Note 1 for definition of L numbers	SECTION OR TABLE WITH GROUP CONTENT DETAILS
	FROM	TO		
46 to 47	—	—	Group Not Used	—
48	MS4	MS1	L1	Section 6-4-2 Table 6-3
49 to 54	—	—	Group Not Used	—
55	Facility Power	Filter	Customer Supplied	Section 6-4-8 Table 6-9
56	—	—	Group Not Used	—
57	Filter	MS4	Customer Supplied	Section 6-4-8 Table 6-9
58 to 68	—	—	Group Not Used	—
69	MDP	MS5	Customer Supplied	Section 6-4-8 Table 6-9
70	SCC	SCC2	L3	Section 6-4-5 Table 6-6
71	MS5 in SCC	MS1	L1/L2	Section 6-4-4 Table 6-5
72	MS8 in SCC	MS1	L1/L2	Section 6-4-4 Table 6-5
73 See Note 2	Backup Water	SCC	Customer Supplied	Section 6-4-8 Table 6-9
74	—	—	Group Not Used	—
75	MS5 in SCC	PP1	L2	Section 6-4-3 Table 6-4
76	MS8 in SCC	PP1	L2	Section 6-4-3 Table 6-4
77	PD1	RF COMMON GROUND STUD	Customer Supplied	Section 6-4-8 Table 6-9
78	RF COMMON GROUND STUD	MS1 GROUND STUD	L1	Section 6-4-2 Table 6-3
79	OW1	PA1	L4	Section 6-4-6 Table 6-7
80	PA1	MG2	L4	Section 6-4-6 Table 6-7
81	MDP	PP1	L2	Section 6-4-3 Table 6-4
82	SCC	SCC2	L3	Section 6-4-5 Table 6-6
83 to 85	—	—	Group Not Used	—
Note 1 Interconnects LOCATION is defined as the following: L1 Within Magnet Room between components L2 Between Penetration Panel (PP1) and components in Equipment Room L1/L2 Between Magnet Room and Equipment Room components through PP1 L3 Between components within Equipment Room L4 Between Operator Workspace (OW) and PP1 L5 Between OW and components in Equipment Room 2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).				
(Continued)				

6-4-11 Cable Group & Location Cross Reference (Continued)

TABLE 6-11 (Continued)
LOCATION & CABLE GROUP CROSS REFERENCE TO RUNS DETAILS

CABLE GROUP #	BETWEEN UNITS		LOCATION See Note 1 for definition of L numbers	SECTION OR TABLE WITH GROUP CONTENT DETAILS
	FROM	TO		
86 See Note 2	SCC	MG2	L1/L2	Section 6-4-4 Table 6-5
87	PCC	MR1	L3	Section 6-4-5 Table 6-6
88	PCC	MG2	L1/L2	Section 6-4-4 Table 6-5
89	—	—	Group Not Used	—
90	PD1	OW1	L5	Section 6-4-7 Table 6-8
91	MR2	OW1	L5	Section 6-4-7 Table 6-8
92	PP1	OW1	L4	Section 6-4-6 Table 6-7
93	MDP	MSM1 & MSM3	L3	Section 6-4-5 Table 6-6
94	MSM1	SCC	L3	Section 6-4-5 Table 6-6
95	MSM1	PP1	L2	Section 6-4-3 Table 6-4
96	MSM1	MR2	L3	Section 6-4-5 Table 6-6
97	MSM1 or MSM4 (Option)	MSM4	Customer Supplied	Section 6-4-8 Table 6-9
98	MSM1	MSM2	L5	Section 6-4-5 Table 6-6
99	MSM4	MDP	L3	Section 6-4-5 Table 6-6
100	MDP	MUX	L3	Section 6-4-5 Table 6-6
101	MSM4	MUX	Customer Supplied	Section 6-4-8 Table 6-9
102	Phone Line Connection	MSM4	Customer Supplied	Section 6-4-8 Table 6-9
103	MSM1	OW1	L5	Section 6-4-7 Table 6-8
104	MUX	OW1 InSite Modem	Customer Supplied	Section 6-4-8 Table 6-9
105	SCC2	MG2	L1/L2	Section 6-4-4 Table 6-5
106	MSM3	MUX	L3	Section 6-4-5 Table 6-6
107	SCC	MR2	L3	Section 6-4-5 Table 6-6
Note 1 Interconnects LOCATION is defined as the following: L1 Within Magnet Room between components L2 Between Penetration Panel (PP1) and components in Equipment Room L1/L2 Between Magnet Room and Equipment Room components through PP1 L3 Between components within Equipment Room L4 Between Operator Workspace (OW) and PP1 L5 Between OW and components in Equipment Room 2 This Group contains water lines which shall be routed separate from electrical lines (i.e. power & signal).				

6-5 CONTRACTOR FURNISHED COMPONENTS

Table 6-12 lists contractor furnished components and details for connections to the system.



Continuous water cooling is critical for the Shield/Cryo Cooler Compressors and therefore MUST be available 24 hours per day / 7 days per week to maximize proper uninterrupted magnet operation. Water cooling is required immediately upon magnet arrival. The System Cooling Cabinet (SCC) and Main Disconnect Panel (MDP) must be installed and operational prior to magnet arrival.

TABLE 6-12
CONTRACTOR FURNISHED COMPONENTS

ASSOCIATED EQUIPMENT	MATERIAL/LABOR PROVIDED BY CUSTOMER CONTRACTOR
POWER IN MAGNET ROOM	Provide and install power and wall duct for Magnet Rundown Unit. (See Section 5, POWER REQUIREMENTS , for power specifications.)
SYSTEM GROUND	Provide ground cable between RF shielded room common ground point and Power Distribution Unit (PD1). (See Section 5-4-2, SYSTEM GROUND , for cable specifications.)
EQUIPMENT POWER • Low Voltage Step-up Transformer* • Main Disconnect Panel (MDP) • MDP to Power Distribution Unit (PD1) • MDP to System Cooling Cabinet (SCC) • SCC to RCU* • Magnet Rundown Unit • Service Outlet in Magnet Room	Provide and install power, conduits, duct work, receptacle, and/or coverplate as required for each item listed. See Section 5, POWER REQUIREMENTS , for power specifications.
EQUIPMENT POWER • SCC to SCC2	Install provided interconnects. See Section 6-2 POWER INTERCONNECTS and Section 6-4 SYSTEM INTERCONNECTS for interconnects details.
ELECTRICAL WIRING	Required for SCC with Remote Condenser Unit (RCU) optional configuration: Provide and install power wiring between SCC and RCU, refer to Section 6-4-9 System Cooling Cabinet With Remote Condenser Unit Additional Interconnects .
PLUMBING	Water cooling for the Shield/Cryo Cooler Compressor Cabinets supplied by the System Cooling Cabinet (SCC) is required at time of magnet delivery. Install plumbing for all water cooling equipment, refer Section 4, SITE ENVIRONMENT , for customer supplied components & requirements. Refer to Section 4-4 WATER COOLING REQUIREMENTS for backup temporary water cooling for all SCC configurations and specific requirements for Water Cooled SCC and SCC with Remote Condenser Unit (RCU).
Note * Optional Equipment	
(Continued)	

6-5 CONTRACTOR FURNISHED COMPONENTS (Continued)

TABLE 6-12 (Continued)
CONTRACTOR FURNISHED COMPONENTS

ASSOCIATED EQUIPMENT	MATERIAL/LABOR PROVIDED BY CUSTOMER CONTRACTOR
REFRIGERATION	Required for SCC with Remote Condenser Unit (RCU) optional configuration: Provide & install refrigeration lines between SCC and RCU, refer to: • Section 2-7-1 System Cooling Cabinet Subsystem Equipment Table 2-5 for listing of responsibility for the specific installation tasks • Section 6-4-9 System Cooling Cabinet With Remote Condenser Unit Additional Interconnects.
CRYOGENIC VENTING	Provide and install cryogenic vent system. (See Section 4-9, CRYOGENIC VENTING , for vent specifications.)
PENETRATION PANEL MOUNTING HARDWARE	RF shielded room vendor to provide appropriate mounting hardware for GE supplied penetration panel. (See Section 7, RF SHIELDED ROOM .)
RF DOOR SWITCH AND CABLING	RF shielded room vendor to provide and install RF door switches on all RF shielded room doors. All switches must be wired in series. GE supplies a 100 ft (30.5 m) cable from System Cabinet which is terminated with 2 leads. These leads are connected to the set of switches. Switches must be in the open position when RF door is open but closed when door is closed. (See Section 7, RF SHIELDED ROOM .)
MAGNET ROOM FLOOR	RF shielded room vendor or customer contractor to provide and install filler around magnet mounting plate, refer to Section 7-7 MAGNET MOUNTING REQUIREMENTS INSIDE RF SHIELDED ROOM .